

Partial Differentiation, exact differentials and Thermodynamics : ①

(Book : Heat and Thermodynamics,
Zemansky and Dittman)

Equation of state :

An equation of equilibrium, connecting the thermodynamic co-ordinates, called an equation of state, is a mathematical function relating the appropriate thermodynamic co-ordinates of a system in equilibrium.

Every Thermodynamic system has its own equation of state, although some of them can be so complicated that they cannot be expressed in terms of simple mathematical functions.

An equation of state expresses the individual peculiarities of one system as compared to another, and must therefore be determined either by experiment or molecular theory.

A general theory like thermodynamics, based on general laws of nature, is incapable of generating an equation of state of any system.

It is not a theoretical deduction, but an experimental addition to thermodynamics.

Equation of state:

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A general equation of state, with variables x, y, z (and so on) can be expressed as

$$f(x, y, z) = 0 \quad (1)$$

↳ Some function (multivariable).

Examples:

Consider an ideal gas obeying Boyle's law, Charles law, Pressure law.

The Thermodynamic variables are pressure (P), volume (V) and temperature (T).

Equation of state here is the ideal gas equation

↳ ideal gas constant

$$PV - nRT = 0$$

↳ number of moles.

$$\rightarrow f(P, V, T) = 0$$

Real gas $\rightarrow$ van der Waals equation.

$$\left(P + \frac{a}{v^2}\right)(v - b) - RT = 0.$$

$$\rightarrow g(P, v, T) = 0$$

$a, b \rightarrow$ constants.

$v = \frac{V}{n} \rightarrow$ molar volume.

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The example of a gas at equilibrium is a special case of a general class of systems that can be described in terms of the three thermodynamic co-ordinates P, V, T .

There are many other types of systems that can be described by these co-ordinates or some more generalized ones.

Let us call a generalized system to be " X, Y, Z system".

These systems can be described by an equation of state at equilibrium.

$$F(X, Y, Z) = 0$$

An equation of state is basically a constraint between the variables.

If we consider two of the three variables to be independent, the third one will be limited by some equation.

$$Z = f(X, Y)$$

Now then, from the fundamentals of partial differential calculus, we can write.

$$dZ = \left(\frac{\partial Z}{\partial X} \right)_Y dX + \left(\frac{\partial Z}{\partial Y} \right)_X dY$$

$\left(\frac{\partial Z}{\partial X} \right)_Y \rightarrow$ Infinitesimal change in Z for such change in X , with Y kept constant.

Mathematical Theorems:

There are two simple theorems in partial differential calculus that are used very often.

Consider an equation of state

$$F(X, Y, Z) = 0.$$

X can be considered as a function of Y and Z .

Therefore

$$dX = \left(\frac{\partial X}{\partial Y} \right)_Z dY + \left(\frac{\partial X}{\partial Z} \right)_Y dZ$$

Also, we can similarly write

$$dY = \left(\frac{\partial Y}{\partial X} \right)_Z dX + \left(\frac{\partial Y}{\partial Z} \right)_X dZ$$

Substituting the second equation in the first, we get

$$dX = \left(\frac{\partial X}{\partial Y} \right)_Z \left[\left(\frac{\partial Y}{\partial X} \right)_Z dX + \left(\frac{\partial Y}{\partial Z} \right)_X dZ \right] + \left(\frac{\partial X}{\partial Z} \right)_Y dZ$$

$$\cancel{dX} / \cancel{dX} \Rightarrow$$

We thus have

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$$dx = \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial x}\right)_z dx + \left[\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x + \left(\frac{\partial x}{\partial z}\right)_y \right] dz$$

Of these three co-ordinates, only two can be independent. Let us choose x and z to be independent.

If thus $dz = 0$ and x not equal to zero, we have

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial x}\right)_z = 1$$

$\Rightarrow$

$$\left(\frac{\partial x}{\partial y}\right)_z = 1 / \left(\frac{\partial y}{\partial x}\right)_z$$

$\Rightarrow$ True for all variables

Now if, $dz \neq 0$ and $dx = 0$, then

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x + \left(\frac{\partial x}{\partial z}\right)_y = 0$$

$$\Rightarrow \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x = - \left(\frac{\partial x}{\partial z}\right)_y = - \frac{1}{\left(\frac{\partial z}{\partial x}\right)_y}$$

$$\Rightarrow \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

P, V, T system.

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$$\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V = -1$$
$$\left(\frac{\partial P}{\partial V}\right)_T = 1 / \left(\frac{\partial V}{\partial P}\right)_T$$

Volume expansibility β and isothermal compressibility κ can be defined as

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P \quad \left[\beta = \text{rate of change of volume (} V \text{) with temp. (} T \text{) per unit volume} \Rightarrow \beta = \left(\frac{\partial V}{\partial T}\right)_P / V \right]$$
$$\kappa = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T \quad \left[\kappa = \frac{\text{Stress}}{\text{Strain}} = \left(-\frac{\partial V}{V} \frac{\partial P}{\partial P}\right)_T \right. \\ \left. (\text{Temp constant}) \right]$$

Therefore, by using the theorems, we find

$$\left(\frac{\partial P}{\partial T}\right)_V = - \left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

$$\Rightarrow \left(\frac{\partial P}{\partial T}\right)_V = - \left(-\frac{1}{\kappa V}\right) (\beta V)$$

$$\Rightarrow \boxed{\left(\frac{\partial P}{\partial T}\right)_V = \frac{\beta}{\kappa}}$$

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An infinitesimal change in pressure may now be expressed as

$$dP = \frac{\beta}{\alpha} \left(\frac{\partial P}{\partial T} \right)_V dT + \left(\frac{\partial P}{\partial V} \right)_T dV$$

$$\Rightarrow \boxed{dP = \frac{\beta}{\alpha} dT - \frac{1}{\alpha V} dV}$$

At constant volume

$$\Rightarrow \int_{P_i}^{P_f} dP = \frac{\beta}{\alpha} \int_{T_i}^{T_f} dT$$

$$\Rightarrow (P_f - P_i) = \frac{\beta}{\alpha} (T_f - T_i).$$

At constant ~~pressure~~ temp.

$$dP = -\frac{1}{\alpha V} dV$$

$$\Rightarrow P_f - P_i = -\frac{1}{\alpha} \int_{V_i}^{V_f} \frac{dV}{V}$$

$$\Rightarrow P_f - P_i = \frac{1}{\alpha} \ln \left(\frac{V_i}{V_f} \right).$$

Stretched wire

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A wire can be thought as an one dimensional system. A sufficiently complete thermodynamic description of a wire is given in terms of three co-ordinates

1. The tension in the wire F .
2. The length of a wire L .
3. The temperature of a wire T .

Hook's Law : $F = -K(L - L_0)$.
→ Hooke's constant
→ Length at zero tension.

$$dL = \left(\frac{\partial L}{\partial T} \right)_F dT + \left(\frac{\partial L}{\partial F} \right)_T dF$$

Now linear expansibility (α) = $\frac{\text{Change in length per unit length}}{\text{change in temp.}}$

$$\Rightarrow \boxed{\alpha = \frac{1}{L} \left(\frac{\partial L}{\partial T} \right)_F}$$

Young's modulus (Y) = $\frac{\text{Change in tension / unit area}}{\text{Change in length / unit length}}$

$$= \left(\frac{\partial F/A}{\partial L/L} \right)_T$$

$$\Rightarrow \boxed{Y = \frac{L}{A} \left(\frac{\partial F}{\partial L} \right)_T}$$

Thus using the theorems we find

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$$\left(\frac{\partial F}{\partial T}\right)_L = -\alpha AY$$

Therefore we have

$$dF = \left(\frac{\partial F}{\partial L}\right)_T dL + \left(\frac{\partial F}{\partial T}\right)_L dT$$

$$\Rightarrow dF = AY \frac{dL}{L} + \alpha AY dT$$

Other such systems are

- a) Dielectric slabs
- b) paramagnetic rods
- c) Electrochemical cell.

} Find the thermodynamic representation
(home work
not assignment)